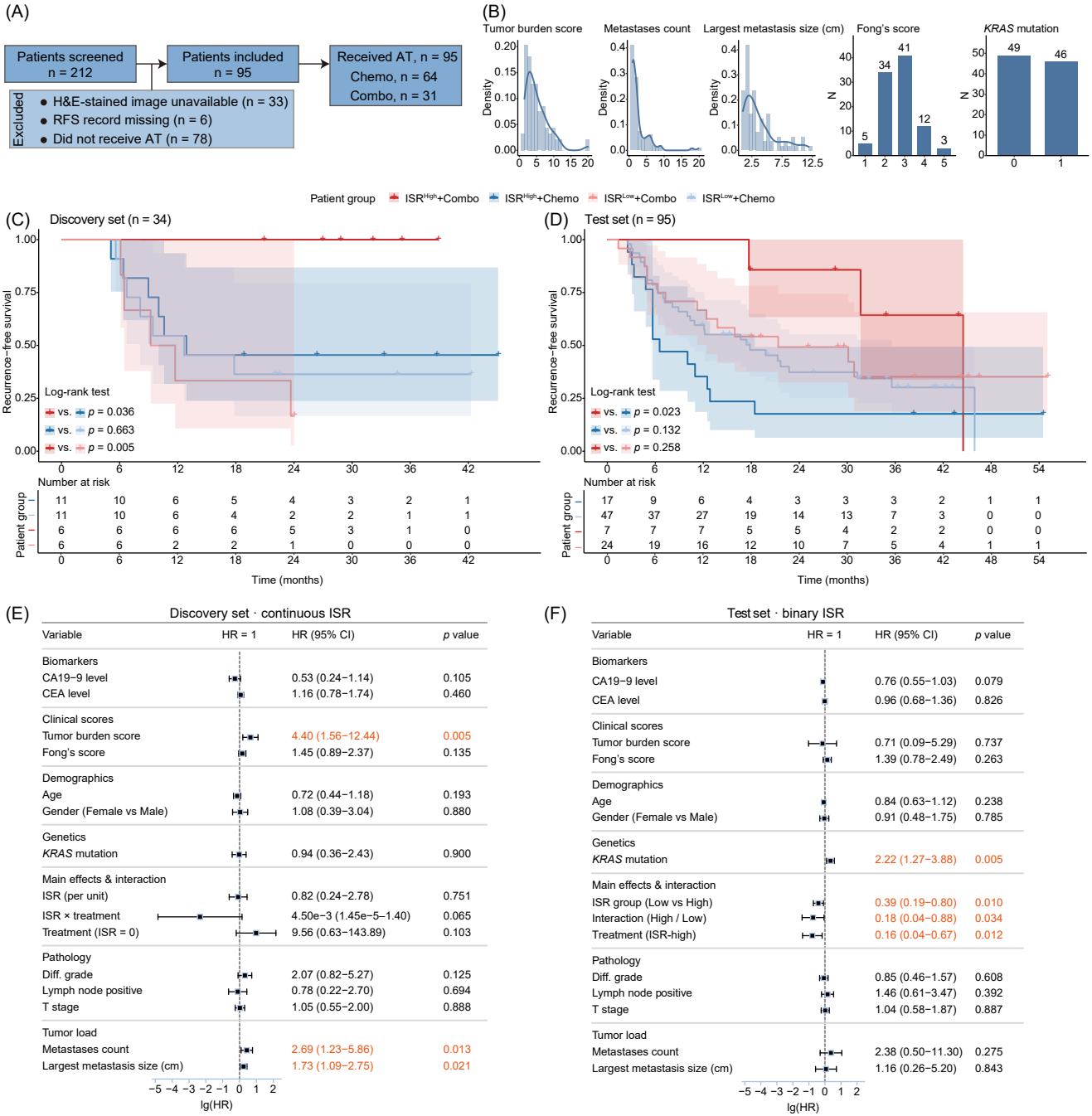

(A)
Patients screened
n = 212
Patients included
n = 95
Received AT, n = 95
Chemo, n = 64
Combo, n = 31
Excluded
H&E-stained image unavailable (n = 33)
RFS record missing (n = 6)
Did not receive AT (n = 78)
(B)
Tumor burden score
Metastases count
Largest metastasis size (cm)
Fong's score
KRAS mutation
Density
N
Patient group
ISR^High+Combo
ISR^High+Chemo
ISR^Low+Combo
ISR^Low+Chemo
(C)
Discovery set (n = 34)
Recurrence−free survival
Log-rank test
vs. p = 0.036
vs. p = 0.663
vs. p = 0.005
Number at risk
Time (months)
(D)
Test set (n = 95)
Log-rank test
vs. p = 0.023
vs. p = 0.132
vs. p = 0.258
(E)
Discovery set · continuous ISR
Variable
HR = 1
HR (95% CI)
p value
Biomarkers
CA19−9 level 0.53 (0.24−1.14) 0.105
CEA level 1.16 (0.78−1.74) 0.460
Clinical scores
Tumor burden score 4.40 (1.56−12.44) 0.005
Fong's score 1.45 (0.89−2.37) 0.135
Demographics
Age 0.72 (0.44−1.18) 0.193
Gender (Female vs Male) 1.08 (0.39−3.04) 0.880
Genetics
KRAS mutation 0.94 (0.36−2.43) 0.900
Main effects & interaction
ISR (per unit) 0.82 (0.24−2.78) 0.751
ISR × treatment 4.50e−3 (1.45e−5−1.40) 0.065
Treatment (ISR = 0) 9.56 (0.63−143.89) 0.103
Pathology
Diff. grade 2.07 (0.82−5.27) 0.125
Lymph node positive 0.78 (0.22−2.70) 0.694
T stage 1.05 (0.55−2.00) 0.888
Tumor load
Metastases count 2.69 (1.23−5.86) 0.013
Largest metastasis size (cm) 1.73 (1.09−2.75) 0.021
lg(HR)
(F)
Test set · binary ISR
Variable
HR = 1
HR (95% CI)
p value
Biomarkers
CA19−9 level 0.76 (0.55−1.03) 0.079
CEA level 0.96 (0.68−1.36) 0.826
Clinical scores
Tumor burden score 0.71 (0.09−5.29) 0.737
Fong's score 1.39 (0.78−2.49) 0.263
Demographics
Age 0.84 (0.63−1.12) 0.238
Gender (Female vs Male) 0.91 (0.48−1.75) 0.785
Genetics
KRAS mutation 2.22 (1.27−3.88) 0.005
Main effects & interaction
ISR group (Low vs High) 0.39 (0.19−0.80) 0.010
Interaction (High / Low) 0.18 (0.04−0.88) 0.034
Treatment (ISR-high) 0.16 (0.04−0.67) 0.012
Pathology
Diff. grade 0.85 (0.46−1.57) 0.608
Lymph node positive 1.46 (0.61−3.47) 0.392
T stage 1.04 (0.58−1.87) 0.887
Tumor load
Metastases count 2.38 (0.50−11.30) 0.275
Largest metastasis size (cm) 1.16 (0.26−5.20) 0.843
lg(HR)